This craft based activity is fun and many useful observations and questions can arise from it.

Students make two simple models:

- One is a paper ladder to represent a short length of the DNA molecule and is used to show the helical structure and illustrate the notion of a gene(s).
- The second is a paper chain which represents a chromosome. It is used to show how the long, thin DNA molecule shortens and condenses as it coils up to form a chromosome.

You will need:

- lined a4 paper
- scrap paper
- scissors
- sticky tape.

Ladder instructions

Cut one A4 sheet in half lengthways.

Take one piece and fold in half, lengthways. If the paper is lined it's helpful to have the margin on the outside so it is visible.

Start at one end and cut out a block two lines deep, cutting from the fold along a line and up to and along the margin.

Move along the sheet a distance of two lines. Leave these intact and cut out a block of the next two lines, up to the margin.

Continue up the length of the sheet, removing about 8 blocks.

Opening the sheet will reveal a ladder.

Paper chain instructions

Simply make a long paper chain. Get students to make sections and then join them together to make one long chain.
Use scrap paper or coloured paper, each link should be about 1-5 cm wide

Show students two or three 3-D animations of DNA molecules. Mute the video so students are not overwhelmed with keywords.

Using the ladders

Below are some suggestions for questions and things to do with the ladders.

1. What does the ladder represent?

2. Students can colour sections, three or four rungs long, to represent individual genes. Students then have a go at twisting their ladders. How well does this show the structure of DNA compared to other models and diagrams they have seen?

3. The rungs of the ladder usually collapse so the helix is difficult to see – how can they solve this? (Students may suggest things like using short strips of sticky tape to reinforce each rung and joined two or more ladders together).

4. Some 3-D animations show the DNA molecule 'unzipping' and two new strands forming. How could they use their ladder models to illustrate this?

Using the paper chains

Below are some suggestions for questions and things to do with the chains.

1. What does the chain represent?

2. Which parts might be said to model genes?

3. Are the ladder and the chain actually showing the same thing?

4. Which is a more accurate model of the double helix? Why?

5. Now, twist the chains. (Hopefully, each link is stuck together firmly). What happens to the chain? Show a 3-D animation which illustrates this.

Further questions

1. If you had a giant model cell, where would you put the ladder or chain you have made?

2. When would you coil up the chain to show the formation of a chromosome?

3. When would you open up your ladder to create two copies of the DNA molecule?

Suggested answers

Using the ladders

1. What does the ladder represent?
A short length of DNA molecule; DNA with a couple of genes; double helix structure of DNA

4. Some 3-D animations show the DNA molecule 'unzipping' and two new strands forming.
How could they use their ladder models to illustrate this?
They could cut them up the middle of the rungs and new T-shaped pieces of paper added to make two new ladders

Using the paper chains

1. What does the chain represent?
A single chromosome and how it becomes more visible under microscope when it coils up.

2. Which parts might be said to model genes?
Each link of chain.

3. Are the ladder and the chain actually showing the same thing?
Ladder represents a small section of the chain.

4. Which is a more accurate model of the double helix? Why?
Ladder shows the 'backbone' of DNA molecule and cross-pieces. Chain has links which pass through one another.

5. Now, twist the chains. (Hopefully, each link is stuck together firmly). What happens to the chain?
Shortens; thickens

Questions

1. If you had a giant model cell, where would you put the ladder or chain you have made? Inside the nucleus

2. When would you coil up the chain to show the formation of a chromosome?
When cell is preparing to divide.

3. When would you open up your ladder to create two copies of the DNA molecule?
During process of cell division when chromosome is making a copy of itself